

Source: theorem P_n for all
natural n

Base case: prove P_0

Assumption: inductive
hypothesis P_n

Inference: prove P of n plus
1 from IH

Conclusion: by induction P
holds for all n

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graph TD; A[Source: theorem P_n for all natural n] --> B[Base case: prove P_0]; B --> C[Assumption: inductive hypothesis P_n]; B --> D[Conclusion: by induction P holds for all n]; C --> E[Inference: prove P of n plus 1 from IH]; E --> D;
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The diagram illustrates the steps of mathematical induction. It begins with a source statement (red box) that a theorem P_n holds for all natural n . This leads to the base case (green box) where P_0 is proven. From the base case, the process branches: one path leads directly to the conclusion (purple box) that P holds for all n , while the other path enters the inductive step. The inductive step (orange box) assumes P_n and uses it to prove P_{n+1} (blue box). Finally, the conclusion (purple box) is reached, stating that P holds for all n by induction.